



PowerFactory 2024

DIGSILENT Fully Rated Converter Wind Turbine Templates

POWER SYSTEM SOLUTIONS
MADE IN GERMANY

POWERFACTORY

PowerFactory 2024

Technical Reference

Digsilent Fully Rated Converter Wind Turbine Templates

Digsilent FullyRatedConv WTG 2.5MW 50Hz,
Digsilent FullyRatedConv WTG 2.5MW 60Hz

PF2024

POWER SYSTEM SOLUTIONS
MADE IN GERMANY

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1 Introduction

This document describes the fully rated wind turbine generator (WTG) template *DlgSILENT FullyRatedConv WTG 2.5MW* that is delivered with *PowerFactory* and explains how to use and configure the model.

The template represents only the grid side converter, it does not include the mechanical part of the rotor and the aerodynamics. This level of detail is for most studies sufficient because the frequency VSC converter decouples the rotor and generator from the grid. This assumption includes also an optimal copper resistor in the DC circuit.

The fully rated WTG model could be used for:

- Load Flow Studies
- Stability Studies (RMS)
- Transient Studies (EMT)
- Balanced and unbalanced simulations (however control is implemented for the positive sequence only)

For dynamic simulation is a variable step size possible. The minimum step size or the fix step size should be lower than 5 ms.

The fully rated WTG model is available for 50 Hz and 60 Hz applications in the global template library:

- DlgSILENT FullyRatedConv WTG 2.5MW 50Hz
- DlgSILENT FullyRatedConv WTG 2.5MW 60Hz

Each template contains also three transformer types for 10, 20 and 30 kV. The model from the template is at the beginning equipped with the 20 kV transformer.

2 How to use this Template

PowerFactory contains pre-defined templates of dynamic models for different generic wind turbines, photovoltaic generators and a battery energy storing system. Using the templates is simple, just click on the *General detailed templates* button () and select the desired template in the following window. Thereafter you could place the model in the single line diagram. For this you have to click one time in the single line diagram, this shows the model connected to the mouse pointer a second click creates a new model of the selected template in the single line diagram. Each dynamic model consists of some elements in the single line diagram (mostly a terminal with a generator) plus a composite model, which is located in the recording grid.

After you placed all models in the single line diagram you have to close the selection window with the templates, this terminates the template inserting.

Do not delete the terminal to which the generator is connected - this would destroy the measurement points of all measurement devices (located in the corresponding composite model).

You could delete a placed model directly by pressing the undo button () or you could delete the model also manually. If you want to delete a model manually you have to mind that the model consists of visible elements in the single line diagram and also of a composite model which is only visible in the Data Manager.

3 Model Description

The fully rated WTG is in the single line diagram represented by a static generator. The models of the controllers are collected in the composite model *FullyRatedConv Control*. This composite model could be found either through the link *Plant Model* on the Basic Data page of the WTG or with the Data Manger in the used grid.

The composite model *FullyRatedConv Control* is created from the frame definition *WTG FRC Frame incl Current Ctrl* this frame is shown in Figure 3.1.

The single slots are briefly described in Table 3.1. Each slot could be filled with either a DSL-model, a measurement device or a *PowerFactory* element, such as a static generator. All measurement devices are connected either to the terminal or to the cubicle, which connects the generator with the terminal. This approach ensures that all measurement devices are correctly connected after entering the model using the template.

Table 3.1: Frame description

Slot Name	Description	Needed Type
ActivePowerReduction	Reduces the power in case of electrical over frequency	DSL-Model
Current Controller	Calculates from current reference a voltage signal for the static generator	DSL-Model
Generator	Static Generator as grid element	*.ElmGenstat
Iac	AC-Current measurement device	*.StaImea
PLL	Fast voltage angle measurement device	*.ElmPhi
PQ	Active and reactive power measurement device	*.StaPqmea
PQ Control	Controls active and reactive power through the rotor current	DSL-Model
Slow FrequMeas	Frequency measurement for over frequency power reduction	*.ElmPhi
Vac	AC-voltage measurement device	*.StaVmea

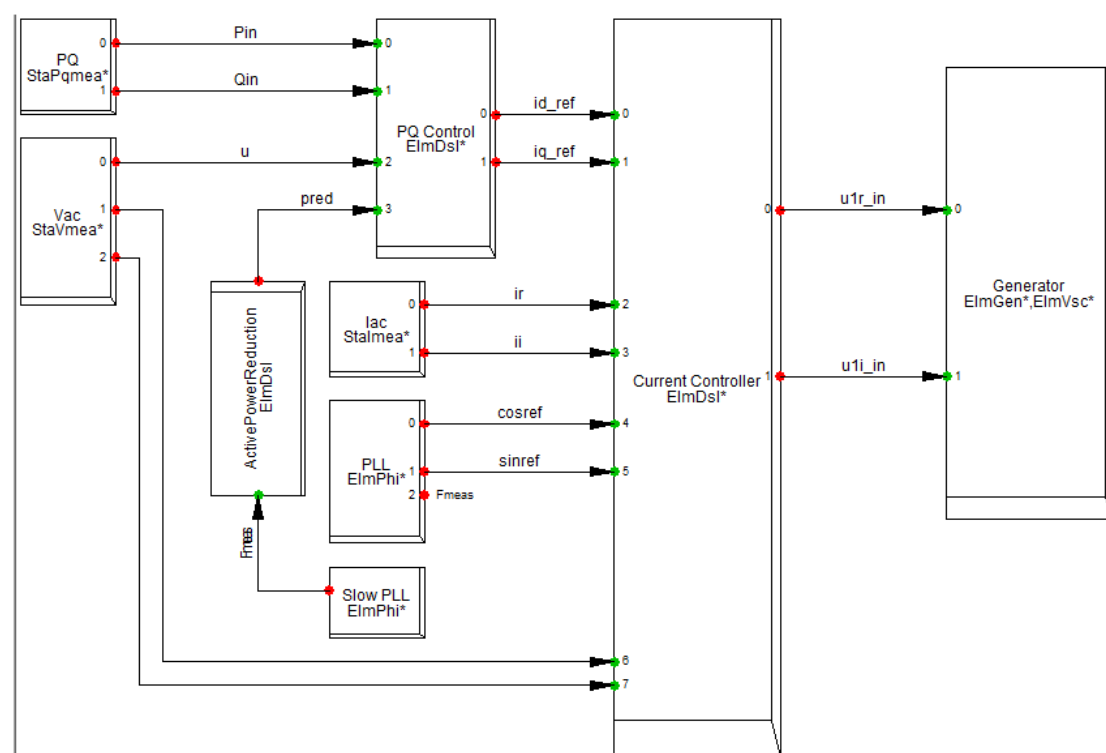


Figure 3.1: Frame Definition WTG FRC Frame incl Current Ctrl (*.BlkDef)

4 How to Customize the Model

You will find in this chapter a description how you could configure the model from the template according to your specific needs. The configuration is sorted to the calculation functions because there are fewer configurations needed for load flow calculation than for a dynamic simulation.

4.1 Load Flow

Changing the number of parallel machines

For grid impact studies of a wind farm it is often sufficient to model the wind farm with one aggregated machine. The generator from the template represents only one machine, but this could be changed very fast in the generator element on the basic data page. For load flow studies are no further changes needed.

Changing the nominal power

The nominal power is for the load flow model only defined by the generator data. If you want to change the nominal power you have to change this on the basic data page and on the load flow page of the static generator. This change affects only the machine you edit right now.

Changing the nominal AC voltage

You could also change the nominal voltage of the machine if your machine has a different nominal grid side voltage. The only modification you have to do is changing the nominal voltage

of the terminal to which the static generator is connected.

4.2 Short Circuit (VDE / IEC / ANSI / complete)

The short circuit behaviour of the model is also completely determined by the element settings. There are therefore no changes in the dynamic model parts necessary. It is important to note that the standard short circuit methods could not give precise results for a controlled wind turbine. For more precise results are dynamic simulations needed. The model of the fully rated WTG is based on the static generator. The static generator allows an individual configuration of the fault contribution for short circuit calculations according to VDC, IEC and complete method. Please read the technical documentation of the static generator for more details [2].

4.3 Dynamic Simulation (RMS/EMT)

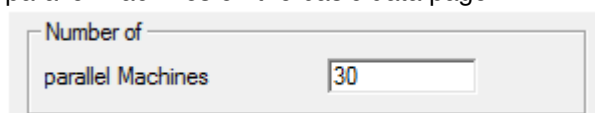
The main purpose of the model created with the template is of course the dynamic simulation. The great advantage of using a template for creating the wind turbine model is that all measurement points are already correct defined and all common models are already configured.

The model is generic and therefore completely open - you could configure it according to your needs. Following you will find a short manual for changing the number of parallel machines, for changing the rated power and for changing some protection settings.

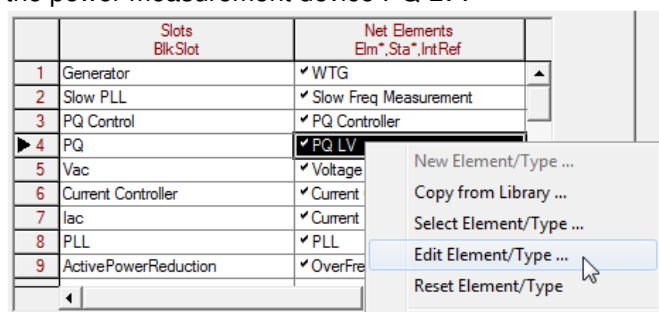
Changing the number of parallel machines

To change the number of parallel machines you have to follow the next two steps:

1. Open the WTG in the single line diagram with a double click and change the number of parallel machines on the basic data page:



2. Open the composite model *FullyRatedConv Control*. To find the correct composite model (if you have more than one) you could double click on the WTG in the single line diagram. On the basic data page is a link to the composite model (Plant Model). In the composite model you have to open (double click or right click and *Edit Element/Type..*) the power measurement device *PQ LV*:



The rated power has to be set to:

$$p_{rating} = n_{parallel} \cdot p_{DFIGrated}$$

Example: 10 parallel machines and a rated active power of 2 MW per machine results in 20 MW.

If you added a step-up transformer you have to change here also the number of parallel devices.

Changing the rated power of the model

Please check if there is already a fully rated WTG model with the needed rated power in the template library before you change the rated power of the fully rated WTG model by hand. If there is no fitting template you have to follow the steps below:

To change the rated power of the WTG to simulate for example a 3 MW WTG instead of the given 2.5 MW you have to follow step 1 and 2. You should have also the amount of parallel machines in mind. The rated power of the WTG is stored in the element data.

1. If you want to change the nominal power you have to change this on the basic data page and on the load flow page of the static generator. This change affects only the machine you edit right now.
2. Open the composite model *FullyRatedConv Control*. To find the correct composite model (if you have more than one) you could double click on the static generator in the single line diagram. On the basic data page is a link to the composite model:



Here you have to change the power measurement *PQ LV*; you have to enter the active power of the WTG times the number of parallel machines. In case of 30 parallel machines and 3 MW you have to enter here 90 MW

Please not that you have also to match the used transformer type.

Changing the nominal AC voltage

You could also change the nominal voltage of the machine if your machine has a different nominal grid side voltage. The only modification you have to do is changing the nominal voltage of the terminal to which the static generator is connected.

Changing the reactive power support during faults

The fully rated wind generator could support the voltage during a voltage dip according to the Transmission Code 2007 and also according to the SDLWindV [1]. The two characteristics are shown in Figure 4.1.

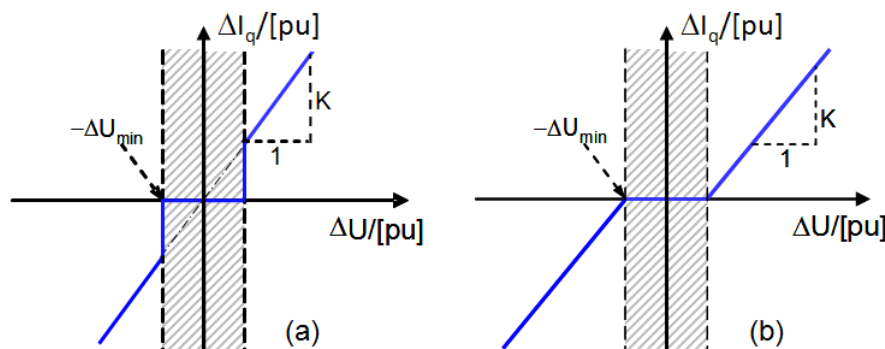


Figure 4.1: Reactive Power Support (a) Transmission Code 2007 and (b) SDLWindV [1]

The settings for the reactive power support could be changed in the *PQ Controller* block. This common model is part of the composite model *FullyRatedConv Control*. To find the correct composite model (if you have more than one) you could double click on the static generator in the single line diagram. On the basic data page is a link to the composite model:



Open then the *PQ Controller*. The parameter i_{EEG} changes between the characteristic (a) if it is 0 and characteristic (b) if the parameter is 1. The voltage deadband could be changed with the parameter ΔU and the gain (indicated with K in Figure 4.1) could be changed with the parameter $K_{\Delta U}$.

If characteristic (a) is selected it is also possible to configure the speed of the active power recovery. This could be done with the parameter *ramp*.

Changing active power reduction during over frequency

The WTG could also reduce the active power during over frequency conditions (for this is also the slow PLL needed). The behaviour of this active power reduction could be configured in the block *OverFrequPowerReduction*. This common model is part of the composite model *FullyRatedConv Control*. To find the correct composite model (if you have more than one) you could double click on the static generator in the single line diagram. On the basic data page is a link to the composite model:

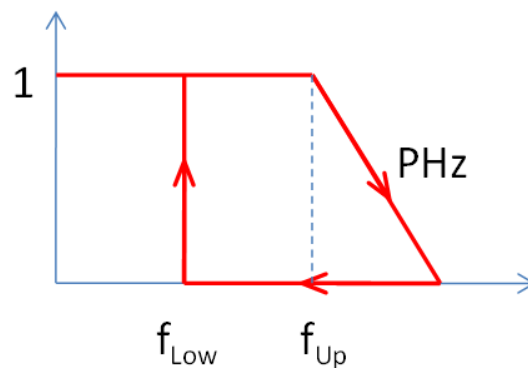


Figure 4.2: Active Power Reduction during Over Frequency

The implemented characteristic is shown in Figure 4.2. There are six parameters which could be changed:

Table 4.1: Frame description

Parameter	Description
fUp	Frequency which triggers the active power reduction (higher than fLow)
fLow	Frequency which ends the active power reduction
PHz	Gradient of the active power reduction (proportional to over frequency)
Tfilter	PT1 filter for frequency measurement
negGrad	Gradient limitation for active power reduction
posGrad	Gradient limitation for active power increase

5 References

- [1] Ordinance on System Services by Wind Energy Plants SDLWindV, 2009.
- [2] *Technical Reference "Static Generator"*. DIgSILENT GmbH, Gomaringen, Germany, 2019.

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